

SAWYER COUNTY AP, WI, USA

WMO: 726508

Lat: 46.026N

Lon: 91.444W

Elev: 1204

StdP: 14.07

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 94973

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.6	-13.1	-25.8	1.4	-17.9	-20.2	1.9	-11.1	17.8	22.1	16.0	22.4	0.7	200	0.552

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.0	87.9	71.0	84.2	69.0	81.8	68.0	74.3	83.3	72.5	80.9	70.5	78.5	8.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.8	123.1	79.2	69.8	114.7	77.5	67.5	105.9	76.1	38.6	83.2	36.9	80.6	35.1	78.1	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	15.1	12.6	-26.6	92.4	3.9	4.1	-29.4	95.3	-31.7	97.7	-33.8	100.0	-36.7	103.0
			-26.8	77.7	3.8	2.4	-29.6	79.4	-31.8	80.8	-33.9	82.1	-36.6	83.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.0	12.8	16.2	28.5	41.8	53.8	63.0	67.9	65.2	57.9	45.3	31.9	18.3		
	DBStd	21.69	13.38	13.00	12.78	9.87	8.96	7.21	6.38	6.14	8.84	9.13	10.24	12.46		
	HDD50	4881	1153	947	675	283	59	3	0	0	25	207	546	983		
	HDD65	8719	1618	1367	1133	698	366	119	43	73	247	615	992	1448		
	CDD50	1965	0	0	7	37	177	394	553	471	262	60	4	0		
	CDD65	327	0	0	0	1	19	59	131	79	35	3	0	0		
	CDH74	3774	0	0	2	24	268	726	1542	853	327	32	0	0		
	CDH80	1017	0	0	0	3	74	178	492	199	69	2	0	0		
	WSAvg	5.1	4.6	5.0	5.4	6.2	6.0	5.0	4.4	4.2	4.9	5.3	5.1	4.8		
	PrecAvg	32.8	0.9	1.0	1.5	2.7	3.6	4.2	4.7	4.3	3.8	3.2	1.9	1.2		
(15) Precipitation	PrecMax	47.0	1.5	2.1	3.6	7.3	5.3	6.9	10.4	7.6	6.9	6.3	5.0	2.1		
	PrecMin	22.6	0.2	0.2	0.6	0.9	1.2	1.5	1.8	1.1	1.0	1.0	0.4	0.3		
	PrecStd	5.9	0.4	0.5	0.7	1.5	1.1	1.4	2.2	2.0	1.6	1.6	1.1	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.4	50.4	67.3	77.4	87.0	89.1	93.3	89.8	86.2	78.7	63.5	46.7		
		MCWB	36.5	43.4	56.7	58.5	66.9	71.7	73.3	72.9	70.7	64.7	52.8	41.8		
	2%	DB	38.2	43.7	59.3	71.4	81.2	84.4	89.1	84.6	81.5	72.0	57.0	41.5		
		MCWB	34.2	38.6	50.7	53.8	63.5	68.5	71.5	69.7	68.3	59.8	49.8	37.3		
	5%	DB	34.7	38.3	52.7	65.8	75.5	81.5	84.9	81.7	78.0	66.0	52.3	37.4		
		MCWB	32.4	34.1	45.1	50.4	60.5	66.9	69.9	68.3	66.3	56.4	45.3	34.5		
	10%	DB	31.4	34.4	46.4	60.9	71.5	78.6	81.9	79.0	73.6	61.4	47.6	35.0		
		MCWB	29.0	31.1	40.1	47.8	57.4	64.7	69.1	66.8	63.7	53.5	42.1	32.8		
	0.4%	WB	37.7	44.0	58.7	61.2	71.6	75.3	78.0	75.7	73.6	69.0	55.8	44.3		
		MCDB	41.8	48.3	65.0	72.6	81.2	84.8	87.2	83.8	81.0	75.1	60.2	46.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.8	39.0	51.3	55.7	66.3	72.1	74.9	73.3	70.7	61.7	50.6	38.1		
		MCDB	37.4	43.5	56.3	67.1	76.5	81.2	84.2	81.4	79.0	69.3	56.8	40.7		
	5%	WB	32.4	34.6	45.6	51.9	63.0	69.2	73.0	71.1	68.2	57.9	46.0	35.0		
		MCDB	34.9	38.1	52.3	63.1	72.1	77.6	81.9	78.2	75.4	64.6	51.0	37.3		
	10%	WB	29.2	31.4	40.4	48.8	59.8	66.9	70.9	68.8	65.7	54.2	42.4	32.7		
		MCDB	31.2	34.0	46.4	60.1	68.7	74.7	79.0	76.1	72.5	60.6	47.2	35.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.6	21.2	21.7	23.1	25.0	23.6	24.0	23.6	23.1	19.7	16.2	16.0		
		MCDBR	18.4	21.8	26.4	33.1	31.0	27.2	27.0	26.5	25.7	27.1	24.3	17.3		
		MCWBR	15.4	16.8	17.2	18.7	15.9	14.3	13.2	14.1	14.4	16.7	17.0	14.2		
		MCDBR	17.3	18.7	23.9	28.4	25.0	22.2	22.7	22.1	21.4	22.1	21.8	15.4		
	5% WB	MCWBR	14.6	14.6	16.5	17.8	14.2	13.3	12.8	13.3	13.2	15.1	16.2	12.9		
(40) Clear-Sky Solar Irradiance	taub		0.247	0.274	0.308	0.344	0.375	0.374	0.369	0.377	0.358	0.312	0.286	0.263		
	taud		2.405	2.332	2.369	2.401	2.350	2.376	2.418	2.408	2.444	2.560	2.523	2.430		
	Ebn at Noon		279	289	293	290	283	283	283	276	272	273	261	258		
	Edh at Noon		22	29	33	35	38	37	36	35	30	23	20	19		
(44) All-Sky Solar Radiation	RadAvg		467	811	1181	1435	1685	1843	1929	1613	1238	765	453	324		
	RadStd		49	71	106	149	125	136	78	109	75	89	51	37		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (45 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page